Machine Learning Fundamentals Guide

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make predictions.

Types of ML: Supervised Learning, Unsupervised Learning, Reinforcement Learning.

ML Workflow: Data Collection → Data Cleaning → EDA → Feature Engineering → Train/Test Split → Training → Evaluation → Deployment.

Important Concepts: Features, Target Variable, Training Data, Testing Data, Normalization, Standardization, Cross Validation.

Regression predicts continuous values. Classification predicts categories. Clustering groups similar data points.

Overfitting occurs when a model memorizes training data. Underfitting occurs when a model cannot learn patterns.

Popular Algorithms: Linear Regression, Logistic Regression, Decision Tree, Random Forest, K-Means, XGBoost.

Machine Learning Interview Questions

1. What is Machine Learning?
2. Difference between AI, ML and Deep Learning?
3. What is supervised learning?
4. What is unsupervised learning?
5. What is reinforcement learning?
6. What is a feature?
7. What is a target variable?
8. What is overfitting?
9. What is underfitting?
10. Difference between regression and classification?
11. What is cross validation?
12. What is feature engineering?
13. What is normalization?
14. What is standardization?
15. What is Random Forest?
16. What is Logistic Regression?
17. What is K-Means Clustering?
18. What is Precision?
19. What is Recall?
20. What is F1 Score?
21. What is ROC-AUC?
22. What is Ensemble Learning?
23. What is Gradient Descent?
24. What is Deep Learning?